

Project Design Phase

Problem - Solution Fit Template

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem Statement:

Farmers, especially in rural India, often select crops based on guesswork or informal advice rather than scientific analysis of their soil nutrients and climate conditions. This leads to suboptimal yield, wasted resources, and financial loss.

Solution:

OptiCrop uses a Machine Learning classification model (Random Forest) trained on real agricultural data to analyze 7 key parameters — Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall — and instantly recommend the most suitable crop, replacing guesswork with a data-driven decision.

Problem Aspect	How OptiCrop Solves It
No scientific guidance available to farmers	Provides instant ML-based crop recommendation from soil and climate data
Risk of choosing the wrong crop and wasting a season	Confidence score and top-3 suggestions help farmer make an informed choice
Limited access to agricultural experts in rural areas	Available anytime on any device via a simple, free web app
Complex agricultural data is hard for a farmer to interpret	Translates 7 raw soil/climate numbers into one clear crop name instantly